$$\lambda = \frac{2\pi\hbar}{p}$$

because the value of the function is unchanged if we add $\frac{2\pi\hbar}{p}$ to the variable *x*:

$$e^{\frac{ip(x+\frac{2\pi\hbar}{p})}{\hbar}} = e^{\frac{ipx}{\hbar}} e^{2\pi i} = e^{\frac{ipx}{\hbar}}.$$

Let's pause for a moment to discuss the importance of this connection between momentum and wavelength. It's not just important: in many ways, it is the relationship that defined twentieth-century physics. Over the last hundred years, physicists have primarily been concerned with uncovering the laws of the microscopic world. This has meant figuring out how objects are built out of smaller objects. The examples are obvious: molecules are made from atoms; atoms from electrons and nuclei; nuclei from protons and neutrons. These subnuclear particles are constructed out of quarks and gluons. And the game goes on as scientists search for ever smaller and more hidden entities.

All of these objects are too small to see with the best optical microscopes, let alone the naked eye. The reason is not just that our eyes are insufficiently sensitive. The more important fact is that eyes and optical microscopes are sensitive to the visible spectrum, which comprises wavelengths at least a few thousand times longer than the size of an atom. As a rule, you can't resolve objects much smaller than the wavelength you're using to look at them. For this reason, the story of twentieth-century physics was in large part a quest for smaller and smaller wavelengths of light—or any other kind of wave. In Lecture 10, we will discover that light of a given wavelength is composed of photons whose momentum is related to the wavelength by exactly the relation